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DEPARTMENT OF COMPUTER SCIENCE ENGINEERING (AI&ML)

MAJOR PROJECT STAGE-1, REVIEW-1 REPORT

ROLE-BASED MULTI-AGENT LLMs for Transparent Decision-Making

*A simulated decision-making committee of Advocate, Skeptic, Compliance-Checker and
Adjudicator agents – built with CrewAI / AutoGen*

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1. Abstract

This project builds a role-based multi-agent LLM system for transparent binary decisions in loan evaluation and academic admissions. A custom Flask/Python two-round debate engine coordinates four roles: **Advocate** presents supporting evidence, **Skeptic** challenges risks and weaknesses, **Compliance Checker** reviews policy and fairness concerns, and **Adjudicator** produces Approved or Rejected with a reason, explanation, and confidence score. The Gemini 3.5 API is the primary backend; Mock AI Mode supports testing without an API connection. Two synthetic datasets are used: 1,000 loan records and 1,000 academic-admission records, with no real student data. JSON audit logs and SQLite store agent outputs, debate history, compliance findings, confidence scores, and final decisions.

Keywords: Multi-Agent Systems, LLMs, Explainable AI, Role-Based Debate, Bias Detection, Auditability, Loan Evaluation, Academic Admissions

2. Literature Survey

The literature reviewed falls into two groups: the foundations of multi-agent debate and orchestration frameworks, and research on bias, fairness and risk in agent-based systems.

2.1 Foundations of Multi-Agent Debate

Reference	Key Contribution and Relevance
Du et al. (2023) arXiv:2305.14325	Introduced multi-agent debate, where several LLM instances critique each other over multiple rounds, improving factual accuracy and reasoning compared with single-agent prompting.
Wu et al. (2024) AutoGen, COLM	Presented AutoGen, an open-source framework for building multi-agent LLM applications with customizable agents and flexible conversation patterns. It is one of the two frameworks used in this project.
Comparative study (2024) AutoGen vs. CrewAI vs. TaskWeaver	Benchmarked the three leading multi-agent orchestration frameworks on collaborative code-generation tasks, informing the choice of framework for this project.

2.2 Bias, Fairness and Risk in Agent Systems

Reference	Key Contribution and Relevance
Emergent Bias and Fairness in Multi-Agent Decision Systems arXiv:2512.16433	Tested multi-agent debate on the German Credit and Adult Income loan datasets. Found that debate can either amplify or reduce bias depending on agent composition, which directly motivates this project's bias-probing objective.
Emergence of Biased Consensus in Multi-Agent LLM Debates arXiv:2608.02827	Showed that conformity among debating agents can trigger a phase transition into collective bias, and that agent heterogeneity helps suppress it.
Weidinger et al. (2022) ACM FAccT	Proposed a taxonomy of risks posed by language models, including discrimination and lack of accountability. This grounds the role of the Compliance-Checker agent in this project.

3. Existing System

In current practice, decisions are produced by a single ML or LLM model that maps applicant data directly to a final decision.



Limitations

- Minimal explanation accompanies the final decision.
- No mechanism exists to challenge or verify the decision before it is finalized.
- Bias goes undetected, as there is no structured probing step.
- The process is difficult to audit or hold accountable.

4. Proposed System

The proposed system replaces the single-model pipeline with a committee of four role-based LLM agents that debate each case before a decision is made. It is built with CrewAI / AutoGen and uses structured debate rounds, JSON transcript logging, and bias probes on synthetic datasets.

No.	Agent	Responsibility
1	Advocate	Argues in favor of the applicant / case.
2	Skeptic	Raises counterpoints and challenges weak reasoning.
3	Compliance-Checker	Flags policy or fairness violations.
4	Adjudicator	Synthesizes the debate into a final, justified decision.

5. Technical Stack

Component	Technology	Purpose
Orchestration	CrewAI, AutoGen	Role-based multi-agent coordination
Language Models	Gemini 3.5 API (primary backend)	Powers each agent; Mock AI Mode for testing without API
Programming	Python 3.11	Core implementation language
Data & Logging	Pandas, JSON, SQLite	Data handling; JSON audit logs and debate history storage
Datasets	Synthetic datasets: 1,000 loan and 1,000 admission records	Evaluation and bias probing (no real student data)
Development	VS Code, Git / GitHub	Development and version control

6. Objectives

1. Design a role-based multi-agent debate protocol for decision-making tasks.
2. Implement transparent, structured logging of agent reasoning as JSON transcripts.
3. Evaluate decision quality and consistency against single-agent baselines.
4. Probe the system for bias using synthetic loan and admissions datasets.
5. Assess whether debate improves fairness and explainability over black-box models.

7. Expected Outcomes

No.	Outcome	Description
1	Working prototype	Auditable, structured multi-agent decision-making system
2	Traceable transcripts	JSON logs showing the full reasoning behind every decision
3	Comparative results	Accuracy and consistency: multi-agent versus single-agent
4	Bias analysis report	Where debate reduces or exposes disparate outcomes
5	Reusable framework	Extensible beyond loan and admissions use cases

8. References

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